

Candelabra Lighting

Operator manual — channels, keys, palettes. White keys pick the preset, octave picks the colour, notes are gated.

MIDI channels

Ch	Addresses	Code	Link key follows
1	both candelabras	F0	—
2	candelabra A	F1	—
3	candelabra B	F2	—
4	all four pars	F40	ch 1, 2 or 3
5	pars 1 and 2	F45	ch 1 or 2
6	pars 3 and 4	F46	ch 1 or 3
7	pars 1 and 4, inside	F47	ch 1
8	pars 2 and 3, outside	F48	ch 1

Channels 9-16 free. No individual candle control — the firmware still answers F11-F13 and F21-F23 if you ever want it back.

F0 addresses the CANDLES ONLY. If it also reached the pars, channel 1 would drive them directly and the link key on channel 4 would mean nothing.

Pars are numbered 1-4 left to right across the stage, so a group like inside spans both nodes. Each node knows which it owns.

Link key — C sharp, any octave

Hold it on a par channel and that group follows its source exactly: preset, master, speed, depth, warmth, fade, hue, sat, boost, every frame. Release and the group snaps back to whatever its own channel had become underneath — a handoff, not a reset.

A group listing several sources follows whichever last changed. White keys on the same channel still set that group's own preset while the link key is down; you just won't see it until you let go. A link never survives transport start or stop.

Idle, fade and offsets

Idle — transport stopped, or five seconds of MIDI silence — is the C-1 flame with a gust on it, read from the palette rather than duplicated, so retuning the flame retunes idle.

Fades. CC 5 sets both directions, CC 20 the attack alone, CC 21 the release alone. All three are global, so one envelope anywhere covers the rig. Every target starts at 150ms. Fade is a PRESET crossfade — changing octave under a held key does not trigger it. An interrupted fade now continues from where it was rather than jumping to peak.

A CC only reaches targets on its own channel, so a note-on first pulls in any CC that target has never been sent — from the newest value seen anywhere, or the resting default. An explicit value on that channel always wins.

Hue and sat offsets are sticky by design, and reset on transport start. They do NOT reset at blackout: MIDI carries no clip boundary, so a new clip and a gap inside one look identical. Put a CC 12 and 13 point at 64 at the start of any clip that must not inherit.

Presets — white keys, any octave

C	0	blackout
D	1	steady
E	2	flicker
F	3	gust
G	4	ember
A	5	pulse
B	6	strobe

CC controls

14	M	master fader
2	S	speed
3	D	depth
4	W	warmth
5	T	crossfade time
12	H	hue OFFSET — 64 = no change
13	C	sat OFFSET — 64 = no change
15	B	white boost — 0 flame, 127 max

Colour palettes — octave

Oct	Colour	hue	sat	wrm	ceiling
C-2	deep ember	20	234	255	27
C-1	candle flame	56	133	0	boost 14
C0	red	255	255	0	—
C1	orange	17	255	0	—
C2	yellow	34	255	0	—
C3	green	73	242	0	—
C4	cyan	125	236	0	—
C5	blue	160	255	0	—
C6	violet	186	252	0	—
C7	magenta	223	255	0	—
C8	full white	0	0	0	boost 255

The ceiling is a MULTIPLIER on the master fader, not a replacement. Deep ember stays dim wherever it is played.

Hue wheel

0	red
43	yellow
85	green
128	cyan
170	blue
213	magenta

Serial, into the transmitter at 115200

F<fix> P<preset> M<master> S<speed> D<depth>
W<warmth> T<fade> H<hue> C<sat> B<boost>

Omitted fields keep their value.
Default flicker: F0 P2 M255 S170 D220 W0 C0 T0

Hue and saturation are OFFSETS. CC 12 at 64 leaves the note's palette colour alone; below and above push either way round the wheel, half a turn at each extreme. CC 13 does the same for saturation, clamped rather than wrapped. Set an envelope's resting value to 64, not 0 — 0 means "shift half a wheel", not "no change". Offsets persist across notes like any CC.

White boost. Nothing in the normal colour path lights every emitter at once — RGB is held down when saturation is low, and white switches off as saturation rises. CC 15 blends toward all of them at full. C8 is the same thing on a key.

CC numbers to avoid: 0 and 32 bank select · 1 mod wheel, DAWs reset it · 6 and 38 data entry · 7 volume · 10 and 11 pan and expression · 96-101 RPN/NRPN · 120-127 channel mode. Safe: 14, 15, 20-31.

Candelabra Lighting

Operator manual — the rig, the pars, and the tuning that makes them match the candles.

Signal chain

Ableton -> IAC -> candle_bridge.py -> USB -> transmitter ESP32
-> ESP-NOW broadcast -> every receiver.

Candle receivers drive 3 SK6812 RGBW pixels each, warm white.
Par nodes convert the same packet to DMX512 over a MAX485 and drive two battery pars apiece down a short XLR chain.

One radio system. The pars' own 2.4G wireless DMX is switched off so nothing competes with ESP-NOW.

Par settings — every fixture

Chnd	10CH
addr	001 / 011 / 021 / 031
FAIL	HOLD hold the look if DMX stops
2.4G	OFF keep it out of ESP-NOW's band
IRCL	OFF no stray remote overriding a cue
NODE	NOR not ECO

Set Chnd before the address — changing mode can reset it.
Leave each display parked on its address. Sitting on CL01, AUT0, SOU0 or NANU means that fixture is running its own effects and ignoring DMX entirely.

Node wiring

GPIO 17	DI
GPIO 16	DE and RE, jumpered
VIN	VCC (MAX485 is a 5V part)
GND	GND
A	XLR pin 3
B	XLR pin 2
GND	XLR pin 1

Node presents a FEMALE XLR. 120 ohm terminator across pins 2 and 3 of a male plug, into the last fixture's DMX OUT.
If a fixture ignores everything, swap A and B first.

Per-node constants

	NODE_ID	NUM_PARS	PAR_INDEX	PAR_ADDR
node A	5	2	1, 2	1, 11
node B	6	2	3, 4	21, 31
bench	5	4	1, 2, 3, 4	1, 11, 21, 31

Par colour tuning

A par is not a warm-white pixel. These correct for that, in the node's firmware, after the packet arrives — so the palettes stay identical for candles and pars.

ww	0.603	white die share
wa	0.974	amber share
gt	0.897	green trim
tf	0.241	flame trim on CH1
tc	1.000	colour trim on CH1
dg	0.710	dimmer curve on CH1
ba	0.300	amber share of boost
tk	1.000	trim curve, 0.5-4.0

ww and wa split the candle's single warm-white value across the par's white AND amber emitters. Both scale with (1 - sat).

tf and tc are interpolated by saturation: tf at sat 0, tc at sat 255.
tk bends that blend — raise it and the flame keeps tf further up the saturation range. It matters because the tuned flame sits at sat 133, no longer the low-saturation look the linear blend assumed.

dg lifts the par's crushed low end. A pixel is linear; a cheap fixture collapses everything under CH1=15. Deep ember lands near CH1=10, which is why it went missing before this existed.

ba holds amber back as boost rises. Boost pushes every emitter to full, and amber at full is amber — a tinted white is not white.

Tuning over the air

F90	whiteW
F91	whiteA
F92	greenTrim
F93	trimFlame
F94	trimColour
F95	dimGamma
F96	boostAmber
F97	trimCurve, 0-255 -> 0.5-4.0

Value rides the master field, 0-255 mapped to 0.0-1.0. These set a variable and light nothing, so they are safe mid-look. Not stored — a reboot returns to the compiled defaults above.

Or over the node's own USB: ww wa gt tf tc dg ba tk, ? to read back.
candle_panel.py does either, and copies values back as source.

candelabra_template.mid

Drag into Live for eight named tracks on the right channels, with all ten CC lanes already present. CC 12 and 13 are seeded at 64, their bipolar centre — a lane starting at 0 means shift half a wheel, not leave it alone. Regenerate with make_template.py.

candle_panel.py

Bench remote. Same F-lines the bridge writes, so close the bridge first — one process per port. Mirror to F40 sends the same look to candles and pars at once, which is the match test. Presets store fixture, preset, parameters, ceiling and all eight trims.

Failsafe. Three independent paths land on a lit fixture rather than a dark stage: no note held is a local blackout, transport stop drops to the idle look, and a receiver that hears nothing for 3 seconds falls to a gusting flame on its own. The F99 heartbeat matches no fixture, so it proves the link is alive twice a second without overwriting anything. The candles fall back to a plain flame, the pars to the tuned one — they carry separate copies, so retuning the palette does not follow them.